

PUTTING INTELLIGENCE ON THE MAP

Biological roots

In the words of a well-known historian of science: 'Intelligence as currently and conventionally understood by psychologists is a brashly modern notion' (Daston, 1992: 211). Indeed, Baldwin's comprehensive *Dictionary of Philosophy and Psychology*, first published in 1901, does not have a separate entry for 'intelligence'. The reader is simply referred to the term 'intellect', which is defined as 'the faculty or capacity of knowing'. At this point 'intellect' was clearly part of the technical vocabulary of philosophical psychology and was presumed to refer to a distinct psychological entity, but 'intelligence' was a mere synonym without any specific scientific meaning of its own. Two decades later Charles Spearman, the British pioneer of testing intelligence, was still complaining that: 'Right up to the present day a large number – perhaps the majority – of even the best accredited textbooks on psychology do not so much as mention the word "intelligence" from cover to cover' (Spearman, 1923: 2).

Nevertheless, in the early years of the twentieth century 'intelligence' had suddenly become a scientific buzz word, among a significant number of psychologists. British psychologists in 1910 (Myers et al., 1910), and American psychologists in 1921 (Thorndike et al., 1921), organized extensive discussions about the meaning of 'intelligence', never doubting that the term had a distinct reference and paying little attention to 'intellect'.

What had happened? Was this merely a linguistic change, a turn of verbal fashion without any fundamental significance? Or should we regard the verbal change as a sign that more interesting changes were afoot? One such change is obvious: intelligence tests had been invented in the meantime and by 1921 were certainly very much part of the psychological map. But does this tell us why 'intelligence' suddenly acquired a psychological life of its own? In a sense, yes, the existence of a widely popular technology was certainly crucial in creating a distinct referential domain for the newly fashionable term. But then again, no, because have we not merely pushed the question one step back? Why,

after all, were the new tests called 'intelligence' tests? They could have been called 'intellect tests', or simply 'mental tests', as their precursors had been called for some time (Cattell, 1890). Why the specificity? To call their tests 'intelligence tests', rather than something else, the testers must have had some preconception of what it was they were testing, or at least wanted to test. Where did that preconception come from?

Perhaps it came from Francis Galton, the nineteenth-century figure who inspired the first generation of Anglo-American intelligence testers? But insofar as Galton uses the term 'intelligence' at all he employs it as an everyday term that is generally interchangeable with 'intellect' and requires no careful definition. The term that he uses more systematically in his studies of human differences is 'natural ability', and this he does define explicitly, though his definition disqualifies it as a synonym for twentieth-century test intelligence: 'By natural ability, I mean those qualities of intellect and disposition, which urge and qualify a man to perform acts that lead to reputation. I do not mean capacity without zeal, nor zeal without capacity, nor even a combination of both of them, without an adequate power of doing a great deal of very laborious work' (Galton, 1962: 77). Now, this is not how intelligence testers defined the object of their investigations. Batteries of intelligence tests were not designed to test for 'zeal', and performance on them was certainly supposed to depend on capacity rather than mere laborious work.

The first point to note then about the conceptual shift from nineteenth-century 'general ability' to twentieth-century 'intelligence' concerns the much narrower scope of the latter. That narrowing becomes quite intelligible when we consider the social context within which each of these concepts was deployed. Galton was interested in life accomplishments of the most general sort, professional success and intellectual recognition. The intelligence testers had been given their start and retained their basis in the school system; it was accomplishment, or rather the lack of it, within the relatively narrow confines of that system which was their primary concern. They were employed to assess *capacity* for scholastic achievement, not 'zeal' or hard work, which teachers could quite easily assess for themselves.

It was not in psychology but in biology that the term 'intelligence' was first elevated from lay usage to systematic scientific discourse. This occurs in the context of analyses of animal behaviour by evolutionary biologists. In 1882 a book appeared that even had 'Intelligence' in the title: George Romanes' *Animal Intelligence*, a title later to be duplicated by E.L. Thorndike (1911). In his book Romanes was concerned with the evolution of mind, and of course the question of reasoning in animals was a key element in that concern. Now, reflection on the nature of human reason had a venerable tradition behind it, but in the new context of evolutionary discourse none of this was particularly appropriate. What mattered now was the notion that the possession of reason

was not an all or none matter but a question of *degree*. Contrary to the Aristotelian and Cartesian tradition, humans were no longer alone in possessing reason, they just had more of it. The reasoning that humans shared to some degree with animals was not the same object as the reasoning which distinguished them from animals. For one thing, the necessary link to language was gone; for another, the concept no longer defined common norms but organized ordered differences. But the term 'reason' is too intimately linked to the old conception, so Romanes must substitute a new term to signal the new discursive context. That term is 'intelligence'. After criticizing the traditional definition of reason, he writes:

More correctly, the word reason is used to signify the power of perceiving analogies or ratios. . . . This faculty, however, . . . admits of numberless degrees; and as in the designation of its lower manifestations it sounds somewhat unusual to employ the word reason, I shall in these cases frequently substitute the word 'intelligence'. (Romanes, 1882: 14)

In the closing years of the nineteenth century this usage became firmly established in evolutionary discourse concerning the relative problem-solving capabilities of organisms. Lloyd Morgan's writings, for example, are full of it (Morgan, 1886, 1891, 1892, 1898).

A decade before Romanes opted for 'intelligence' as the label for a faculty of reasoning inherently variable in its development the conceptual niche for such a faculty had already been prepared in Darwin's own writing. In *The Descent of Man* Darwin had made it clear that his theory entailed differences of degree in what he called 'mental powers' between the species, including humans. He writes:

If no organic being excepting man had possessed any mental power, or if his powers had been of a wholly different nature from those of the lower animals, then we should never have been able to convince ourselves that our high faculties had been gradually developed. But it can be clearly shewn that there is no fundamental difference of this kind. We must also admit that there is a much wider interval in mental power between one of the lowest fishes . . . and one of the higher apes, than between an ape and man; yet this immense interval is filled up by numberless gradations. (Darwin, 1981: 35)

In common with so many of his contemporaries Darwin immediately extends the concept of inherently graded mental powers to the differences among human groups. He continues:

Nor is the difference slight . . . in intellect, between a savage who does not use any abstract terms, and a Newton or Shakespeare. Differences of this kind between the highest men of the highest races and the lowest savages are connected by the finest gradations. (*ibid.*)

For Darwin, variability was a fundamental attribute of what he referred to as 'the intellectual and moral faculties of man', and, he adds, 'we have every reason to believe that the variations are inherited' (Darwin *ibid.*: 159). He certainly had the framework that contained a conceptual

space soon to be filled by the category of 'intelligence', though as yet the label was missing.

Had Darwin been less conservative in his use of language he might have made use of two developments that appear to have had a significant effect on later usage. One of these occurred in France earlier in the nineteenth century when Lamarckian evolutionism had led to a rather lively debate about the possible evolution of reasoning (*raisonnement*) in animals. But people were not comfortable with the idea of reason, traditionally the most human of qualities, being perhaps a matter of degree and being at least partially shared with animals. In 1822 Baron Cuvier put the weight of his considerable authority behind the proposal to use *l'intelligence*, rather than reason, in the context of discussions about the quasi-rational accomplishments of animals (R.J. Richards, 1987: 67). This convention appears to have taken hold, giving the term a recognized place in French scientific discourse at a time when it had not yet acquired this status in English. As the conception of human reason became increasingly naturalized in the course of the nineteenth century (Daston, 1992) the term came to refer to human cognition conceived as a multi-layered structure (e.g. Taine, 1872). By the time that Binet entered the discussion at the end of the century the evolutionary link had done its work. What remained was a concept of cognition as essentially layered or graded.

English usage was slow to adopt these French innovations. But a new period began with the publication in 1855 of Herbert Spencer's *Principles of Psychology* (1871), four years before Darwin's *Origin of Species*. Spencer, a dedicated evolutionist, not only announced the vision of Psychology as a biological science, but presented the category of 'intelligence' as a central concept for such a science. 'Intelligence' was the name Spencer gave to a basic process which, he thought, covered both psychological and biological change. The essential point about minds and organisms, for Spencer, was that they had to adapt to their environment. That adaptation was never perfect but entirely a matter of degree. The better the adaptation, the higher the level of intelligence. So the term now describes, not a quality which an act may or may not possess, but a graded series of more and less satisfactory acts. Spencer's universe is one in which every living being is constantly under test, and the adequacy of its response constitutes the level of its intelligence. The specific content of adaptations is relevant only insofar as it permits the construction of a hierarchy of excellence. 'Intelligence' has come to be defined by an inherently comparative concept that refers to an abstract notion of excellence of performance under test.

It is in Spencer's evolutionist philosophy that 'intelligence' re-emerges as a term with a specific theoretical meaning, rather than being simply one of a myriad psychological terms in everyday usage or a mere synonym for the academically respectable term 'intellect'. Intelligence had had a specific meaning in scholastic and Cartesian philosophy, but

the contrast between Spencerian 'intelligence' and its philosophical predecessors could hardly have been more profound. In the philosophical tradition the term was intimately linked to the fundamental distinction between a material and an immaterial realm, being specifically used with reference to the latter. Hence the once common and now archaic but still intelligible notion of a divine intelligence. It was as spiritual beings that humans could be said to possess intelligence. By contrast, Spencer is at great pains to establish the complete unity of the material and the mental characteristics of living forms. Humans have intelligence by virtue of their status as biological beings. This status not only subjected them to the iron law of adaptation, mental as much as material, but placed them in an order of continuous natural gradations from the lowest organism to the man of genius.

The image of the world as an arena in which living beings were constantly tested was not an idiosyncratic invention of Spencer's but was based on common Victorian convictions that were certainly shared by Darwin and Galton. Although the latter produced the paradigmatic contributions that made these convictions scientifically fruitful, Spencer, the philosopher, had his day when scientific usage gradually conformed to his terminology. When the Darwinian students of animal behaviour adopted 'intelligence' as an object for their science they were essentially adopting Spencerian usage. This intelligence was an inherently graded biological characteristic concerned with task performance in the abstract.

From the beginning, the category of biological/evolutionary intelligence was deployed to rationalize the grading of human groups in a hierarchy of excellence. Among the English-language contributors to the Victorian discourse on intelligence the placing of humans at the top of a scale of animal intelligence was always accompanied by a division of humans into higher and lower orders according to this criterion. Invariably, these orders were distinguished by the criterion of race; almost invariably, by the criteria of gender and social class. At the same time as it bridged the ancient chasm between humans and animals the new category of biological intelligence sanctified chasms between different types of humans. The latter could now be seen as the product of a natural order whose authority had assumed the attributes of divinity.

Intelligence became a biological category at a time when biology had become a field for the application of empirical scientific methods. It was only a matter of time before such methods would be extended to the study of biological intelligence. At first, these studies were limited to 'animal intelligence', but the Darwinian perspective in general, and more specifically the Galtonian programme of eugenics, provided a clear licence for extending the investigation of 'intelligence' to humans. Two followers of Galton, his biographer Karl Pearson, and the retired Indian army engineer, Charles Spearman, acted on this licence at about the same time. In 1904 Pearson published a study on the inheritance of human mental characteristics, including 'ability', defined in terms of

degrees of intelligence (Pearson, 1904). In the same year Spearman published his first paper on 'general intelligence' (Spearman, 1904a). Two years later Pearson followed this up with a study 'On the relationship of intelligence to size and shape of head and to other physical and mental characters' (Pearson, 1906), and Spearman went on to devote the rest of his life to the study of 'intelligence'. At this stage neither Pearson nor Spearman had any inkling of what was afoot in Paris. Binet's intelligence tests had nothing to do with the scientific début of the term in the English-language technical literature, although they undoubtedly helped its rapid progress. When, from about 1908 onwards, Anglo-Saxon investigators became aware of Binet's tests, which came with the label 'intelligence' attached, they had no difficulty in assimilating them to an already existing conceptual schema.¹

What modern intelligence was not

Before we return to the relationship between the twentieth-century concept of intelligence and the social practice of intelligence testing we need to look a little more closely at the conceptual work which the modern term 'intelligence' performed. Like every other abstract category label 'intelligence' took its place in a network of categories among which there were certain affinities and distinctions. For an understanding of the semantic significance of each category both affinities and distinctions are important. Affinities can tell us about what is taken for granted in defining a category in terms of one particular polarity, and distinctions provide clues about what is significant for those who use such polarities as a means of orientation. At different historical times and in different cultural settings a wide and variable range of such distinctions has been considered significant, and the meaning of psychological categories has fluctuated accordingly.

What were the distinctions that played a major role in the appeal of modern 'intelligence'? Given the biological origins of the scientific term it comes as no surprise to find that the contrast between intelligence and instinct was often stressed in the work of the late nineteenth-century animal psychologists. 'Instinct and intelligence' was also the title of the first psychological symposium on the modern concept of intelligence (Myers et al., 1910). The salient distinction here is between behaviour that is the result of *individual* adaptability and learning, and behaviour that is based on stereotyped inherited patterns (Of course, different grades of individual adaptability were considered to be inherited too.) The polarity of instinct and intelligence was not simply a morally neutral cognitive framework but clearly embodied a judgement of relative value. Certainly at the human level there was an unmistakable preference of intelligence over instinct and this expressed the superior valuation of the qualities that intelligence represented.

How did intelligence differ from instinct? First, it depended on individual experience; instinctive action was generally thought to owe nothing to individual learning, though it was often highly adaptive. Secondly, intelligent action was flexible and pliable where instinctive action was inflexible and rigid. The distinction had not always been seen in these terms. Older conceptions of instinct had tended to emphasize its 'wisdom' rather than its stereotypy, and classical notions of intellect had generally credited it with a significant supra-individual aspect (see Chapter 2). The tenacity with which the basis for the modern instinct-intelligence distinction was adhered to in the twentieth century, even in the face of awkward empirical observations, makes one suspect that this basis owes more to the culturally sanctioned value of individual flexibility than it does to the demands of scientific work.

But what about the supposed affinity that linked instinct and intelligence as the contrasting poles of a conceptual pair? What did instinct and intelligence share that made it meaningful to compare them within one and the same discursive framework? The answer to this question has already been given in the previous section of this chapter. Once intelligence became a biological category it automatically took its place in a network of such categories that would now help to define its meaning. 'Instinct' was particularly salient because, like intelligence, it addressed the issue of adaptation or fitness.

'Intelligence' was deeply embedded in biological ways of thought and discourse. That is what distinguished it from the much older category of 'intellect' or reason. If 'intelligence' always had connotations of gradation that 'intellect' lacked, this was but one aspect of its membership in a family of biological concepts. The rise to prominence of a quasi-scientific concept of intelligence occurred in the wake of a powerful late nineteenth-century trend to subsume human life under biological categories. Prior to this there had been no need of a concept like the modern 'intelligence', though there had certainly been notions of 'intellect', 'intellectual faculties', and so on.

In one sense 'intelligence' and 'intellect' were actually contraries, for they were the products of opposite impulses. Where 'intelligence' was introduced as part of a discourse based on the continuity between the human and the sub-human, 'intellect' had traditionally been deployed in the course of attempts at explaining the nature of uniquely human attributes. As was noted in Chapter 2, Aristotle had made a basic division between those characteristics of the human *psyche* which it shared with the *psyche* of other living things and those which were peculiar to humanity. The translation of the latter as *intellectus* survived into the early modern period via Aristotelian scholasticism. 'Intellect' was a specifically human attribute, fundamentally different from anything to be found at the non-human level. It was not unusual for this attribute to be identified with the human capacities for logical inference, conceptual thinking and abstraction. Descartes, to whom the

division between human and animal life was of course supremely important, even developed the concept of the 'pure intellect' (also rendered as 'pure understanding') which emphasized the irreducibility of intellect to anything like sensory experience or gradual learning. This notion lived on in the influential Cartesian tradition (see e.g. Malebranche, 1980).

That the new, biological, conception of intelligence was incompatible with ancient usage did not escape the notice of those who had a vested interest in that usage. Erich Wasmann, a German Jesuit, was scathing in his condemnation of the notion of an intelligence common to humans and animals and differentiating between them only in a matter of degree (Wasmann, 1897). Nor was his the voice of a scientific illiterate, for he was a well-known researcher in the field of animal behaviour who had published his empirical observations, notably on the behaviour of ants, in the standard scientific journals of Germany, Austria and the Netherlands. But for a man like Wasmann, steeped as he was in the conceptual distinctions of another age, the very notion of animal intelligence was nonsensical, for it was intelligence which made humans human. Needless to say, his reactionary polemics had no discernible effect on the onward march of the modern concept of intelligence, but they do serve to underline the magnitude of the conceptual change that had occurred in this area.²

There had actually been two major changes. One was the transformation of intelligence into a biological category, which we are concerned with in the present chapter. But this transformation would not have been possible without the radical subjectivization of reason that we noted in Chapter 3.³ The classical construction, going back to Aristotle and beyond, had worked with categories that referred to a rational order of things as well as to the human capacity to understand that order (see Chapter 2). Even at the end of the seventeenth century a faint echo of this idea is still to be found in the writings of John Locke, a philosopher who often points the way to later developments. For when Locke (1959) comes to speak of the freshly minted notion of 'the association of ideas',⁴ he contrasts two ways in which ideas are connected: by 'chance or custom', and by 'natural correspondence'. The former type of connection occurs as a result of the association of ideas, but the latter type is due to the 'excellency of our reason'. The association of ideas often leads to 'unreasonableness' and even madness; that is, a loss of that rationality which is inherently in tune with the natural order of events.

Locke's lingering belief in a human capacity for discerning the rational order of the world begins to seem dubious in the eighteenth century. The kind of scepticism represented by David Hume in particular tends to abolish the distinction between the merely customary and the truly rational and to make the operation of human reason subject to the mechanics of the association of ideas. This paves the way for a purely

instrumental understanding of intellectual activity (see Chapter 3). What reason can do is to find the best means for achieving an individual's ends, not to discover the best ends. Rationality is not something that can be attributed to the world, only to individuals. Things just are the way they are, though people are equipped to make the best of them. In due course this equipment would be attributed to individuals' biological constitution, but for this attribution to be made it was first necessary for people to get used to thinking of the intellect as a piece of equipment, an instrument like an arm or a leg, to be employed in the service of individual ends. Such a notion would have seemed strange indeed to someone accustomed to the classical perspective.

The universal classroom

The break with tradition entailed by the nineteenth-century concept of biological intelligence did not occur only on the level of ideas. The Spencerian image of life as a universe of graded forms expressed an apparently obvious truth in a world transformed by industrial capitalism and colonialism. It also expressed the pious belief that the gross inequalities of that world simply reflected a natural order of things in which rewards were necessarily distributed according to inherent worth or endowment.

Nineteenth-century industrial capitalism not only provided a hospitable environment for the idea of universally graded worth, it also provided the techniques that were to transform this idea into a socially effective technology. Among these techniques were (1) the standardization of human work; (2) the division of work into specialized performances; (3) the subjection of work activity to quantitative measures of output; and (4) the exercise of a regime of discipline designed to separate the content of work activity from its social purpose (workers had to learn to understand and willingly execute tasks that were not embedded in traditional communal activity).

As countries were swept up by the process of industrialization their educational systems adapted to the new conditions of adult life. Textbooks and syllabuses became standardized. The school day was rigidly subdivided into clock-timed 'lessons', each devoted to a different specialized subject matter. Examinations became much more pervasive and formal, and written examinations (which allowed for verifiable quantification of output) came to replace oral examinations.⁵ Not only was schooling organized differently, it also became universal and compulsory. That entailed the extension of the new system of work discipline to parts of the youthful population that had not previously come within its grasp. Problems within the system were now greatly exacerbated.

A major set of problems revolved around the issue of accountability. The rationalization of schooling affected teachers as much as pupils. Their work too became subject to standardized norms. Quantitative measures of output could be used to judge the performance of teachers as readily as to judge the performance of pupils. Indeed, as teachers lost much of their traditional local autonomy to find themselves on the bottom rungs of an ever more intrusive bureaucratic structure they were increasingly held accountable for the results of their pedagogic efforts. At another level, tax payers wanted to be reassured that the large funds now going into public education were being used efficiently in terms of the results achieved.

Unfortunately, the results achieved within the recently rationalized school system often seemed unimpressive when measured against the expectations of ambitious parents, anxious administrators and suspicious tax payers' representatives. One of the results of transforming education into a *system* for the manufacture of pedagogic output was that everyone within the system – pupils, teachers, administrators – was potentially accountable for perceived failures to make the system work efficiently. In that situation one defensive argument was always available to anyone taken to account: responsibility for disappointing results could be displaced from the process of manufacture on to the poor quality of the raw material, the innate endowment of the pupils. Because it was generally recognized that people differed in their suitability for various social tasks, the argument that some were inherently unsuitable for academic work could be deployed as effectively by individual pupils accused of morally reprehensible laziness as by teachers and administrators challenged to account for poor outcomes. A discourse of differential 'brightness', 'cleverness', 'slowness', 'dullness' etc. therefore became an inevitable feature of the new system. As the fashion for giving social problems a biological interpretation reached ever wider sections of the literate population towards the end of the nineteenth century one can detect a mounting tendency to associate the traditional social labels of 'brightness' and so on with notions of biological superiority and inferiority. In the work of Pearson (1904) and Spearman (1904a), referred to above, we find an attempt to give this association a scientific basis.

The recently rationalized educational system of the late nineteenth century not only gave rise to the phenomenon of intellectual defect as a social problem, it also generated the means which made possible the development of a social technology for handling this problem. Without the prior development of a rationalized system of school examinations in the nineteenth century there would have been no technology of intelligence testing in the twentieth. It was the new system of schooling that gave universal validity to the notion that intellectual activity could be treated as a kind of *performance* to be judged against precise standards of achievement. Systematic formal examinations gave practical expression

to this notion. But then, as we have seen, the problem of accounting for the results of this new social practice had to be faced. Who was to be blamed for poor examination performance, and what was to be done about it?

At this point a clear alternative existed. One could adopt a critical attitude to the system that had produced the disappointing results and advocate its change. A few educational reformers took this path. Alternatively, one could accept the existing system and seek to improve the way it functioned. This was the path taken by the psychologists, and in taking it they devised a technology that really was to put their infant discipline on the map, the technology of intelligence testing.

To understand the birth of this technology one has to keep in mind that at the end of the nineteenth century there existed another system of examination, apart from the academic variety. This was the medical version. Individuals were commonly examined for their social fitness by medical experts – another nineteenth-century development which we do not need to pursue here. This meant that there existed a boundary beyond which the competence of the academic examiner was no longer recognized and medical competence took over. Intelligence testing initially took root by taking advantage of favourable conditions at this somewhat blurred boundary. Medical examination was generally employed to determine whether an individual was to be committed to a medically administered institution like an insane asylum. Academic examination was employed to determine the relative success of large numbers of individuals in educational institutions. Universal compulsory education had however created a sizeable category of individuals who, while dismal failures in school, did not appear to be candidates for an asylum. In due course, special school classes and special schools were created for such individuals, mainly because they were a disruptive influence on the regular process of instruction. But how were individuals to be selected for these special institutions? The sole use of ordinary school examinations for this purpose was questionable because they might simply reflect poor motivation, an unfortunate home environment, conflicts with the teacher, and so on. On the other hand, there would often be nothing about these children that suggested the kind of abnormality for which medical intervention would be considered appropriate. Besides, medical examinations in this area tended to be time consuming, costly and unreliable.

The early intelligence tests were invented to fill this gap.⁶ They were a hybrid form that made it possible to play educational and medical practice off against each other by borrowing elements from both. Psychological examiners within the school system mimicked the role of medical experts who had preceded them. In America they also found it expedient to use a quasi-medical language. But the actual technology that was deployed owed more to nineteenth-century education than to nineteenth-century medicine. The most important components of this

technology were, firstly, the comparison of answers to standard examination questions with a norm, and secondly, the use of age-graded cohorts as the foundation for norms of performance. The former practice had been developed in the context of increasingly rigorous school examinations; the latter was familiar from the bureaucratic use of syllabuses that established different expectations of achievement for age-graded school classes. From there it was but a short step to Binet's concept of 'mental level', changed by others to 'mental age', thereby implying that intellectual age differences were essentially quantitative rather than qualitative.

For their content the items of the new test drew partly on existing medical practice, especially at the lower age levels, where a child would be asked to carry out simple requests and give evidence of understanding the meaning of common verbal formulations. Then there were school-like tasks, tests of vocabulary, rote memory and rhyming ability, that were used at somewhat higher age levels. Previous psychological investigation also played some role in the choice of item content. The resulting instrument was a heterogeneous amalgam tailored to a specific practical requirement.

Binet labelled this instrument a scale of 'intelligence' precisely because it was not designed to test a distinct psychological entity. Intelligence, he recognized, was a 'word so vague and so comprehensive' reflecting the fact that 'nearly all the phenomena with which psychology concerns itself are phenomena of intelligence' (Binet and Simon, 1916: 42). The best that Binet can do to characterize the common feature of everything that falls under 'intelligence' is to invoke the category of 'judgement'. But what is judgement? It is 'good sense, practical sense, initiative, the faculty of adapting one's self to circumstances'; it is 'a fundamental faculty, the alteration or the lack of which, is of the utmost importance for practical life' (Binet and Simon, 1916: 42). Hardly the stuff of rigorous scientific definition! But of course that was not what Binet was after. He had been concerned to produce a practical device that reflected at least some of the myriad meanings that might be intended when, in everyday situations, a child was judged to be lacking in intelligence. He insisted that his scale did not really measure an entity called 'intelligence', because its component items were not superimposable like linear surfaces. One might calculate an individual's overall score on the scale, though if one was to avoid 'illusions' one should do so merely as a matter of practical convenience.

But Binet's little device soon took on a life of its own. This did not happen on its home soil in France, where there was no enthusiasm for reducing intelligence to a single number (Schneider, 1992), but in America and Britain where Binet's test was quickly translated, adapted and packaged as the new psychology's great gift to humanity – the IQ test. Its technology was appropriated by a group of professionals – Spearman and Burt in Britain, Goddard, Yerkes and Terman in the US –

whose understanding of 'intelligence' was based on the late nineteenth-century biological model discussed earlier in this chapter. Henceforth an individual's total score on an intelligence test was interpreted as a measurement of a real entity, something singular that a person could have more or less of, like height or weight. Not only was this entity unequally distributed among individuals, it was also unequally distributed among social classes (the special concern of the British psychologists) and races (the special concern of the Americans). Similarly to physical characteristics like height and pigmentation, the entity called intelligence was also assumed to be inherited. Inserted into this conceptual framework intelligence testing became transformed from a purely practical device with limited purposes into a pervasive technology applicable to the entire population.⁷

Although both the French and the Anglo-Saxon investigators collected their empirical data in school settings, their intentions were rather different. Binet and Simon limited themselves to the narrow technical task of assisting the educational authorities in implementing their policy of determining pupils' fitness for different programmes. Those who had been inspired by Galton, however, had only a limited interest in this modest craftsman's task. Enthralled as they were by the grand eugenic vision they also wanted to lay bare the roots of human worthiness in general. The category of intelligence would serve to bridge the gulf between relative academic success among school pupils and relative success in life among social classes, races and individuals.

For the Galtonians academic success and failure were of supreme interest insofar as the classroom could be regarded as a kind of microcosm that reflected a crucial feature of the macrocosm of life. The bridge between the two was established by conceptualizing both of them as essentially arenas for competition. The school became the model of life insofar as it could be seen as an arena in which innate endowment was put to the test. But this became possible only when life itself had come to be seen as an arena in which individuals and groups were constantly being tested for their relative 'fitness'. When Galton proclaimed: 'I look upon social and professional life as a continuous examination' (Galton, 1962: 49), he gave expression to the world view that made the dominant version of the modern intelligence concept intelligible.

Quite early on there were two versions, mundane and grand, of this concept. They had in common the notion of an inherently graded scale of performances, but whereas the mundane version regarded any such scale as specific to time, setting and task, the grand version saw one of these scales (intelligence tests) as an emanation of an absolutely general and immutable quality of the individual (Andersen, 1994). But to entertain the notion of such a quality, and to believe that intelligence test performance might be its quintessential expression, one had to share Galton's perspective on life as a continuous examination.

Psychometric intelligence

In due course IQ testing not only became an industry, it also provided the motor for an entire branch of psychology. That development was greatly fostered by the mass use of intelligence tests in the American Army during the latter stages of World War I.⁸ Once more there was a convergence of professional interest and practical requirements, as there had been in the case of Binet and Simon's work for French education. But this time the professional interests were represented by a far larger and relatively well organized group of psychologists while the pressures of wartime forced the pace of change. The American Army had been a tiny professional force that was forced to expand as rapidly as possible into a conscripted mass army able to hold its own in twentieth-century industrial warfare. That created enormous problems of manpower selection. American psychologists offered to help with these by adapting the recently developed technology of intelligence testing to military use. At first they were able to play on existing fears of the mentally defective to gain bureaucratic acceptance of a technology that had not been designed for use with adults. But very soon their plans became more ambitious and eventually resulted in the testing of more than one and a half million recruits.

In order to achieve that result the psychologists had to modify their technology. Binet's procedures had stayed close to the model of the medical examination in opting for an unhurried one on one interaction of professional tester and child testee and in leaving room for the examiner's judgement when the child's responses were converted into scores. In the American military context such an approach was clearly unworkable. So the tests were administered to large groups of men at a time (often under conditions of extreme discomfort), and a simplistically codable multiple choice format was substituted for experienced professional judgement in arriving at test scores. Military conditions left their mark on intelligence testing and fostered its conversion into a technology for quickly grading large populations in terms of their mental worth. This conversion provided the indispensable basis both for the ideologically charged debates about observed racial differences in IQ and for the mass use of intelligence tests in the educational system during the 1920s and 1930s.

But the military influence went further than this. Strict timing was introduced and answers to test items had to be begun and finished promptly on command. Army authorities greatly valued speed and obedience, and if 'intelligence' was to be assessed in this context the conditions of testing would have to reflect these priorities. Most important of all, the intelligence test scores of recruits had to be validated against officers' ratings of their intelligence in order to secure military acceptance of the new technology. Psychologists had of course encountered an analogous situation when they first tried to develop

instruments for assessing the intellectual level of schoolchildren. If these instruments did not yield results roughly in accord with teachers' ratings no one, including other psychologists, would be convinced by their claim to measure 'intelligence'. Whatever it was that intelligence tests could be said to 'measure' it had to be broadly similar to the understandings prevalent in the specific institutional settings in which large scale validation of the tests had taken place.

But there was also a reciprocal influence which psychologists and their technology exerted on the institutional environments that gave them a base. Insofar as these institutions made any use of what the psychologists produced, the category of 'intelligence', now equated with the results of IQ tests, came to play a bigger role in the way these institutions rationalized their selection policies and decisions. The use of psychological expertise in institutional settings always involved some kind of trade-off and negotiation in which institutional authorities permitted a certain degree of psychological influence on condition that their own priorities were reflected in the psychologists' work. Never was psychological knowledge simply 'applied' in an otherwise passive setting.

Nor were the psychologists merely the passive recipients of institutional influences. Their sense of identity as scientists and their existence as a specific community of experts depended upon two articles of faith: that they were engaged in the investigation of real natural objects and that they possessed the appropriate technical means for generating knowledge about these objects. For early twentieth-century American army and school psychologists the most important natural object was 'intelligence' and IQ tests formed the appropriate technology. The latter conviction was reflected in the (incorrect) belief that statistical correlations among the items of an intelligence test provided unambiguous evidence of the operation of a unitary intellectual power inhabiting each individual. Faith in the existence of intelligence as a unitary natural object was grounded in a whole network of beliefs constituting an ideology of the Social Darwinist type. For at least the first quarter of the twentieth century eugenics was an almost invariable component of this ideology.

Whatever their basis, these convictions provided many psychologists with an ideological backbone that enabled them to resist day to day pressures to modify their procedures in light of concrete institutional requirements. Some 'applied' psychologists, like Walter Dill Scott (Mayrhauser, 1987), were ostracized as being too accommodating by those who regarded themselves as the guardians of the profession's scientific soul. But though the latter were also collaborators up to a point, their ideologically driven stance paid off after the war when American psychology could claim to have established scientific proprietary rights over an entity of great social importance, the national intelligence. As long as hereditarianism, xenophobia, racism and Social

Darwinism were significant ingredients of public life the psychological category of intelligence and its attendant technology assured the new discipline a respected place in the social fabric. As J.M. Cattell, a veteran of American psychology, put it in 1922: 'The army intelligence tests have put psychology on the map of the United States', though he had the wit to add: 'extending in some cases beyond these limits into fairyland' (quoted in Samelson, 1979: 106).

The massive institutional application of intelligence tests perpetuated the split between ideological and practical interests that had emerged when these instruments were applied in a military context. Ideologically, the results of testing were supposed to provide evidence for the existence of a unitary, inherited intellectual power whose unequal distribution would scientifically legitimize political restrictions on immigration, access to educational resources, etc. Practically, testing was supposed to provide the kind of information about individuals that could be used to bring about a better fit between diverse individual characteristics and specific social tasks and expectations, as in vocational guidance or the detection of learning disabilities. Such practical purposes were better served by a multiple rather than a unitary conception of test intelligence (Mayrhauser, 1992). More than any other branch of twentieth-century psychology the area of test intelligence always remained deeply and directly implicated in extra-scientific issues of a political and administrative kind (Harwood, 1983). As a consequence there was always significant social backing both for ideological conviction and for pragmatic considerations among members of the research community. This was likely to perpetuate divisions of opinion about the unitary or multiple nature of intelligence. Moreover, the results of the mathematical technique of factor analysis, which was supposed to resolve such conflicts, themselves depended on non-mathematical decisions that could be influenced by supposedly banished extra-scientific considerations. Unanimity on the nature of intelligence therefore eluded the field.

Matters were further complicated by the fact that ideological influences were historically diverse. In North America the ideological pendulum began to swing away from Galtonian hereditarianism towards a rather simple-minded environmentalism during the second quarter of the century. Psychological research on intelligence followed suit and many studies demonstrating the impact of environmental influences on IQ were conducted (Cravens, 1985). However, it must be emphasized that this development had no very profound influence on the use of the category of intelligence in psychology. Environmentalists generally accepted the IQ as a genuine measure of some significant and unitary entity whose reality they did not seek to question. By shifting the question from the reality of what was essentially a hereditarian construction to its precise genetic status the environmentalists actually helped to legitimize a fundamental component of their opponents' conceptual framework (Samelson, 1975). This prepared the ground for a

sterile debate about the 'heritability' of IQ, a debate that can be kept going for ever if the key conceptual questions about the nature of the IQ and the nature of heritability are never asked (Lewontin 1975, 1985; Schiff and Lewontin, 1986; Wahlsten, 1994).

Historically, the crucial step in the scientific survival of the intelligence concept was the establishment of an indissoluble link between a certain view of intelligence and a specific technology. For the discipline there were undeniable political advantages in espousing the psychometric concept of intelligence. The technology of testing provided psychology with its first significant marketable product and established psychologists as the recognized authorities on the nature of intelligence. This was signified quite early in the famous definition, or rather slogan, that intelligence was whatever intelligence tests measured. One reason why 'the intelligence men' (Fancher, 1985) clung to the IQ as the representation of a real and singular entity must surely have been their considerable professional investment in this notion.

Not that there was any lack of suggestions for ways out of the impasse into which the psychology of intelligence was driven by its tie to the technology of IQ testing. However, the alternative psychologies of intelligence which produced these suggestions all sprang up outside psychology's North American heartland and were easily marginalized. A brief enumeration of these alternative psychologies must begin with Gestalt Psychology, one of whose earliest contributions was Wolfgang Köhler's (1925) study of the intelligence of apes. Among other things, this work emphasized a fundamental distinction between cognitive processes that were essentially 'blind' and mechanical and those that were insightful. A related distinction, between 'reproductive' and 'productive' problem-solving activity, was proposed by another Gestalt psychologist, Max Wertheimer (1945). Psychometric intelligence was not concerned with such subtleties because of its exclusive concern with the final outcome or yield of cognitive processes rather than the manner in which those outcomes had been achieved. Because the data of intelligence tests consisted merely of a collection of 'correct' and 'incorrect' judgements the only kind of theoretical concept that they seemed to require took the form of a hypothetical innate limit on performance, i.e. an ability. Thus there developed a somewhat notorious split between the psychology of cognitive performance and ability on the one hand and the psychology of cognitive process on the other. The two had nothing to say to each other, and until the advent of the computer and 'artificial intelligence' the study of problem solving as a process rather than a performance received relatively little support.

The determination to shake off testing technology's stultifying focus on intellectual yield and to look instead at process was also central to the alternative psychology of intelligence developed by Jean Piaget (1950). As in the case of Gestalt psychology, this led to distinctions among qualitatively distinct manifestations of 'intelligence'. In the long-

buried but recently rediscovered work of Vygotsky (van der Veer and Valsiner, 1991; Wertsch, 1985) one also finds suggestive hints at an alternative psychology of intelligence. These emerge out of an interest in assessing intellectual possibilities rather than limitations. There is also a break with the tradition that defines ability only in terms of what individuals can do on their own, a tradition which informs the received psychology of intelligence, in both theory and practice.

Looking back at the historical development of the received view, one notices a curious feature that calls for a concluding comment. On what grounds have certain procedures been accepted as providing a measure of 'intelligence' rather than some other psychological quality or qualities? Because their results correlated with those of other such tests or with tests of academic achievement? Undoubtedly, but then the same question arises with regard to these other tests, leading us into a historical regress which is not endless. Sooner or later we always end up, not with a score, but with a judgement, a judgement that a certain performance or a certain individual merits the attribute 'intelligent' or its opposite in some degree. The very first attempts at assessing intelligence had to be validated in this way, and even the power of professional expertise could not long protect a conception of intelligence that diverged significantly from such culturally embedded judgements (Carugati, 1990).

Perhaps then it would have been better to approach the psychology of intelligence, not with the *a priori* assumption that intelligence is a quality which individuals possess, but with the observation that intelligence is something attributed to individuals in certain judgement situations (Goodnow, 1984). That would have led to questions about the nature of the situations in which such judgements are made, the expectations and implicit criteria on which such judgements are based, the variation of such judgements with the characteristics of the performance or the person judged, etc. This is a strictly counterfactual speculation, because nothing of the sort happened. But a dash of counterfactual history can sometimes serve to put into sharper relief the things that were always taken for granted in the historical developments that did occur.

Notes

1. For a comprehensive account of these developments and their historical background see Carson (1994).

2. Charles Spearman (1937), whose scientific career depended on taking the objectivity of the modern category of intelligence for granted, could see no incompatibility between this category and scholastic conceptions of an intellectual faculty. Twentieth-century psychologists sometimes engaged in a little historical tourism, returning home with the claim that they had spotted an ancient lineage for concepts that were always in danger of being regarded as 'brashly modern'.

3. This is a change that has continued to engage social philosophers in the twentieth century, often evoking diametrically opposed reactions. The representatives of modernism have welcomed the change (e.g. John Dewey, 1930: 203), while critical theorists have deplored it (e.g. Max Horkheimer, 1947: ch. 1). Dewey explicitly recognized that the modern term 'intelligence' was not just a technical term but stood for a whole world view. The psychologists' reification of the term would seem to be based on an implicit acceptance of this world view.

4. This topic did not appear in the original 1690 version of Locke's *Essay* but was added to a later edition as a kind of afterthought. Contrary to a popular legend in the historiography of modern psychology Locke was not the founder of 'associationism'. I have discussed this at greater length elsewhere (Danziger, 1990b).

5. For an overview see Calhoun (1973: part 2), and Hanson (1993: ch. 7).

6. This story has often been told. The most detailed English-language account of the original invention of this instrument in France is to be found in Theta H. Wolf (1973). The essentials are carefully discussed in R.E. Fancher (1985). For a translation of the primary source, see A. Binet and T. Simon (1916). American developments are discussed in JoAnne Brown (1992); and in Leila Zenderland (1987).

7. 'What was originally a device for diagnosing the defective became a device for hierarchising the normal' (Rose, 1985: 128). For an analysis of the conceptual problems entailed by these developments, see also Gould (1981: chs 5 & 6).

8. This episode in the history of modern psychology has come in for a great deal of scholarly scrutiny, starting with Kevles (1968) and continuing with a landmark study by Samelson (1979). Important recent contributions are those of Mayrhauser (1991) and Carson (1993). I have drawn on these sources in the following paragraphs.

BEHAVIOUR AND LEARNING

The story of twentieth-century academic Psychology is the story of an ultimately unsuccessful struggle against an ever more obvious fragmentation. Intelligence and its testing provided an early example of the discipline's tendency to annex new areas without being able to assimilate them. Psychologists had gained an academic foothold by doing experiments on such topics as sensation, perception and memory. For some time, that remained the respectable core of the discipline, but how test intelligence related to this core was far from clear. It was much easier to annex such a field institutionally than to assimilate it intellectually.

That situation was to be repeated many times over in the course of the twentieth century. Child study, or paedology, as it was known in some countries, was another early example. Originating in joint efforts by physicians and educationists, it became transformed into child psychology, rapidly in the US, more slowly in Europe. But its links to core areas of the discipline remained tenuous at best. The same could be said of educational psychology, another early branch. In the period between the two world wars the discipline sprouted as many arms as Shiva, the Hindu deity. A psychological social psychology challenged its sociological rival, 'personality' and 'motivation' emerged as semi-autonomous fields of research and teaching, industrial psychology flourished, clinical psychology became a reality.

What link was there between these fields, except that they all claimed to be 'psychological'? But did that mean anything beyond a vague sense of a common focus that was based on popular images rather than on solid scientific grounds? Grouping these diverse areas together as branches of one discipline undoubtedly had certain practical advantages. It advanced the cause of professionalization by implying that the more practically oriented branches had a respectable link to basic science, and it legitimized the otherwise esoteric interests of the academics by implying that their work had significant practical applications. But, for the most part, such implications were nothing more than promissory notes to be cashed in at some time in the future. Why should anyone accept these notes, and, more importantly, how could psychologists justify such promises to themselves?